

Q1 Explain the difference between authentication and authorization.

- Authentication is the process of Verifying the identity of a User by checking their Credentials, such as a Username and password. It answers the question "Who are you?"
- Authorization is the process of determining what an authenticated user is allowed to do after successfully logging. It answers the question, "What are you allowed to access or perform?"

Q2 Why should passwords never be stored in plain text?

- Password should never be stored in plain text because anyone who gains access to the database or files can immediately read every user's password. This can lead to unauthorized access, identity theft and compromise of other accounts if user reuse the same passwords. Storing password securely using hashing greatly reduces the risk.

Q3 What is the purpose of password hashing?

- Password hashing converts a password into fixed length encrypted value called a hash using a one-way mathematical algorithm such as SHA-256. Instead of storing the original password, the system stores the hash. During login, the entered password is hashed again and compared with the stored hash. Since hashing is one-way, the original password cannot be easily recovered, Improved Security.

4 How does duplicate login prevention improve security?

→ Duplicate login prevention ensure that a user account can only be logged in from one device or session at a time. This helps prevent unauthorized user from accessing an account using stolen credentials while the legitimate user is already logged in. It also improve session management and reduce the risk of account misuse.

5 Suggest two additional security improvement for your application.

Ans:

1 Encrypt communication using SSL/TLS so that all message and login credentials are protected while traveling across the network.

2 Implement two-factor Authentication (2FA) by requiring users to verify their identity using a one-time-password (OTP) or authentication application in addition to their password.